

Research Article

Alexandr Udeneev and Oleg Bakhteev

Extracting Inductive Bias Using Generative Models

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Abstract: Inductive biases shape how neural networks generalize from limited data, yet encoding them in network connectivity often relies on domain expertise. This paper presents a generative approach that draws on related tasks to discover shared structural constraints. For each task, a generative model approximates a family of hypothesis spaces that admit near-optimal solutions. Intersecting these task-specific families narrows the search to structures compatible with all tasks, automating the discovery of a shared inductive bias (want lossless). Experiments on synthetic tasks demonstrate improved data efficiency and generalization to unseen tasks, supporting the hypothesis that the recovered constraints capture transferable structure.

Keywords: inductive bias, generative model

1 Introduction

Choosing an appropriate inductive bias is crucial for generalization [1]. By constraining the hypothesis space to reflect the underlying task structure, such a bias enables effective learning from limited data. In neural networks, examples of structural inductive biases include local connectivity and spatial weight sharing in convolutional architectures [3], and recurrent state updates with parameters shared across time in recurrent architectures [2]. Suitable parameter-sharing patterns also enforce equivariance to task-relevant transformations [4]. Building on these principles, recent work has explored learning symmetry-inducing parameter-sharing patterns directly from data [5, 6].

Selecting suitable structural biases by hand requires domain expertise, particularly when the relevant properties of a new task are not known in advance [5]. This motivates learning such biases from data. For example, Zhou et al. [6] represent layer weights as the product of a transformation matrix shared across tasks and a vector of task-specific parameters. The shared matrix is meta-learned across tasks to capture transferable parameter-sharing patterns, including those that induce equivariance. For larger layers, the authors use a Kronecker-factorized approximation to reduce memory and computational costs, restricting the search to a structured family of reparameterizations. More generally, a single task may admit multiple near-optimal structures, and its data alone need not identify which constraints will transfer to other tasks. The broader challenge is therefore to identify structural constraints that support learning across an environment of related tasks and generalize to previously unseen tasks from that environment [1].

2 Related Works

TODO

3 Preliminary

TODO

4 Method

TODO

5 Experiments

TODO

6 Conclusion

TODO

References

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